

Q1. What is Machine Learning?

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables systems to learn patterns from data and make decisions without being explicitly programmed for every specific task.

Main Types of Machine Learning:

- * **Supervised Learning:** The model is trained on labeled data (input-output pairs). *Example:* Predicting house prices based on historical sales.

- **Unsupervised Learning:** The model finds hidden patterns or structures in unlabeled data. *Example:* Grouping customers into segments based on purchasing behavior (Clustering).
- **Reinforcement Learning:** An agent learns to make decisions by performing actions in an environment to maximize a reward. *Example:* An AI learning to play chess or driving a self-driving car.

Q2. Supervised Learning: Regression vs. Classification

Supervised Learning involves training a model on a dataset that includes both the input features and the correct output (labels).

Feature	Regression	Classification
Output Type	Continuous / Numerical value.	Categorical / Discrete label.
Goal	To predict a specific quantity.	To predict a class or category.
Example	Predicting the price of a house (\$350,000).	Predicting if an email is "Spam" or "Not Spam".

Q3. Linear Regression Overview

Linear Regression is a statistical method used to model the relationship between a dependent variable and one or more independent variables by fitting a linear equation to observed data.

The Equation :-

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \epsilon$$

- **Y (Dependent Variable):** The target value you want to predict (e.g., house price).
- **β_0 (Intercept):** The value of Y when all independent variables (X) are zero.
- **β_1, β_2 (Coefficients):** These represent the change in Y for every one-unit change in the respective X variable.
- **ϵ (Error Term):** The difference between actual and predicted values.

Q4. Evaluation Metrics

These metrics help determine how well the regression model is performing:

- **SSE (Sum of Squared Errors):** The sum of the squares of the residuals (deviations predicted from actual empirical values of data).
- **MSE (Mean Squared Error):** The average of the squared differences between predicted and actual values.
- **RMSE (Root Mean Squared Error):** The square root of MSE; it provides the error in the same units as the target variable.
- **R^2 Score (Coefficient of Determination):** Represents the proportion of variance for the target variable that's explained by the independent variables.
- **Adjusted R^2 :** A modified version of R^2 that adjusts for the number of predictors in the model, penalizing the addition of unnecessary variables.

Q5. Machine Learning Workflow

The standard workflow follows these steps:

1. **Data Collection:** Gathering the raw data required for the problem.
2. **Data Preprocessing:** Cleaning data, handling missing values, and encoding.
3. **Exploratory Data Analysis (EDA):** Visualizing data to find patterns and correlations.
4. **Feature Selection:** Choosing the most relevant variables for the model.
5. **Model Training:** Splitting data into training/testing sets and training the algorithm.
6. **Model Evaluation:** Using metrics (like RMSE, R^2) to check accuracy .